

Lab #7: Keras MLP for Multiclass Classification

Objective

Implement a **Multi-Layer Perceptron (MLP)** using Keras/TensorFlow for a **multiclass classification** problem and analyze the effect of different activation functions, optimizers, and model configurations on classification performance.

Problem Statement

Select a suitable real-world **multiclass classification dataset**, such as:

- Iris species classification
- Handwritten digit classification
- Wine-quality/category classification
- Human activity recognition
- Customer or product category classification
- Any other multiclass dataset approved by the instructor

Using **Keras/TensorFlow**, develop an MLP-based classification model to predict the class/category of each observation.

The selected dataset must contain **three or more classes**.

1. Dataset Preparation

Students must:

1. Load and explore the selected dataset.
2. Provide a brief description of the dataset and classification problem.
3. Identify the input features and target/class variable.
4. Determine the number of classes.
5. Analyze the class distribution.
6. Handle missing values, if any.
7. Encode categorical features, if required.
8. Encode the target variable appropriately.

9. Perform suitable feature scaling/normalization.
 10. Split the dataset into training and testing sets.
 11. Use a suitable validation strategy during training.
 12. Justify the preprocessing techniques used.
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2. MLP Model Development

Design an MLP architecture containing:

- Input layer
- **At least two hidden layers**
- Output layer with **one neuron for each class**

For multiclass classification, use an appropriate output activation function such as:

Softmax

Implement the model using **Keras/TensorFlow**.

Students must report:

- Number of input features
 - Number of classes
 - Number of neurons in each hidden layer
 - Activation function used in each layer
 - Output activation function
 - Optimizer
 - Learning rate, if modified
 - Batch size
 - Number of epochs
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3. Experiment A – Hidden-Layer Activation Functions

Implement and compare at least **three activation functions**, such as:

- ReLU
- Sigmoid
- Tanh

Keep other important hyperparameters reasonably consistent while comparing the activation functions.

For each experiment, record:

- Training accuracy
- Validation accuracy
- Training loss
- Validation loss
- Test accuracy
- Precision
- Recall
- F1-score

Students must explain why the selected activation functions may behave differently for the chosen multiclass classification problem.

4. Experiment B – Optimizer Comparison

Select and compare at least **three optimizers**, such as:

1. SGD
2. Adam
3. RMSprop

Keep the dataset split and basic MLP architecture consistent.

Compare the effect of the optimizers on:

- Convergence speed
- Training loss
- Validation loss
- Classification accuracy
- Generalization performance

Students must justify why the selected optimizers are suitable for training an MLP.

5. Model Training

Train each model using the **same training/testing split**.

Use suitable values for:

- Epochs

- Batch size
- Optimizer
- Learning rate
- Validation split

Record the training and validation performance for every experiment.

Plot:

- Training Accuracy vs. Epoch
- Validation Accuracy vs. Epoch
- Training Loss vs. Epoch
- Validation Loss vs. Epoch

Use these plots to identify:

- Overfitting
- Underfitting
- Poor convergence
- Stable convergence
- Differences in convergence between optimizers/activation functions

6. Model Evaluation

Evaluate each final model on the **unseen test dataset** using suitable multiclass classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Use **macro-average** or **weighted-average** metrics where appropriate and clearly mention which averaging method was used.

Students must visualize and interpret the confusion matrix.

Identify:

- Which class is predicted most accurately?
- Which class is most frequently misclassified?
- Which pair of classes is difficult for the model to distinguish?

- Possible reasons for misclassification.
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7. Comparison Table

Prepare a comparison table containing the experimental results.

Experiment	Activation	Optimizer	Test Accuracy	Precision	Recall	F1-Score
Model 1	ReLU	Adam				
Model 2	Sigmoid	Adam				
Model 3	Tanh	Adam				
Model 4	ReLU	SGD				
Model 5	ReLU	RMSprop				

Students may conduct additional experiments if required.

8. Analysis and Interpretation

Based on the experimental results, answer the following:

1. Which activation function performed best? Why?
 2. Which optimizer performed best? Why?
 3. Which activation-function and optimizer combination produced the best overall performance?
 4. Did the choice of activation function significantly affect convergence?
 5. Did the choice of optimizer affect the speed of convergence?
 6. Which model achieved the best validation performance?
 7. Which model achieved the best test performance?
 8. Is there any significant difference between training and testing performance?
 9. Does the learning curve indicate overfitting or underfitting?
 10. Which classes were most frequently misclassified?
 11. What could be the possible reasons for these misclassifications?
 12. What changes could improve the classification performance?
 13. Which model would you select as the **final model**, and why?
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9. Final Model Selection

After completing all experiments, select the **best-performing MLP model**.

The final model selection must consider more than accuracy.

Students should consider:

- Test accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Training/validation curves
- Convergence behavior
- Model complexity
- Generalization performance

Provide a clear justification for the final model.

10. Deliverables

Submit a **Google Colab notebook** containing:

- Dataset description and source
- Problem definition
- Dataset exploration
- Class distribution analysis
- Data preprocessing
- Feature encoding/scaling
- MLP architecture
- Keras/TensorFlow implementation
- Experiments with different activation functions
- Experiments with different optimizers
- Training/validation plots
- Confusion matrices
- Evaluation metrics
- Comparison table
- Final model selection
- Justification and interpretation
- Conclusion

Important Requirements

Students must **not simply change the activation function or optimizer and report accuracy.**

The experiment must include **justification, comparison, validation, and interpretation.**

Students must explain how the characteristics of the selected dataset—such as:

- Number of classes
- Class imbalance
- Feature distribution
- Non-linearity
- Noise
- Overlapping classes
- Number of training samples

can influence the performance of the MLP.

All experiments must use the same dataset split and evaluation procedure to ensure a fair comparison.

Expected Learning Outcome

After completing this laboratory, students should be able to:

- Build an MLP for multiclass classification using Keras/TensorFlow.
- Apply appropriate preprocessing for classification data.
- Select suitable activation functions and output layers.
- Compare different optimizers.
- Evaluate multiclass classification models using multiple performance metrics.
- Interpret confusion matrices and learning curves.
- Select and justify an appropriate final MLP model based on experimental evidence.